Qn 7	**Answer**	**Marks**
(a)	When a constant DC used, the magnetic field from primary coil remains constant, no changing in magnetic flux linkage through the secondary coil. Hence, by Faraday's law, no induced e.m.f. in the secondary coil.	**[1]** **[1]** **[1]**
(b)(i)	Power delivered to the industrial consumer $= 0.99 \times 4200 \times 10^3$ W $= 4.158 \times 10^6$ W $= 4.16 \times 10^6$ W	 **[1]**
(b)(ii)	current in secondary coil of the transformer U = $\sqrt{(4.16 \times 10^6) / 30)} = 372.3$ A current in primary coil of the transformer U = 372.3 / 275 X 11 = 14.89 A resistance in supply cable = $P/I^2 =$ (0.010 x 4,200,000) / (14.89 x 14.89) = 189 Ω	**[1]** **[1]** **[1]**
(b)(ii)	Voltage across secondary coil of the transformer T = 4,200,000 / 14.89 = 282 kV	**[1]** **[1]**
	Max Marks	**9**

Qn 8	Answer	Marks